

CHAPTER - 3

Lenskart E-Commerce Database Management System

3.1 Introduction

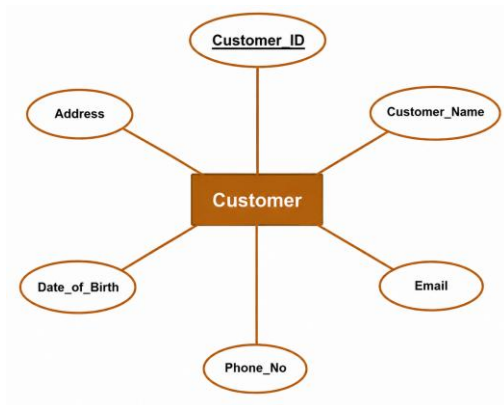
Lenskart is one of India's leading eyewear retailers, offering spectacles, sunglasses, contact lenses, and eye-care products through online and offline channels. The Lenskart Analysis System is designed to manage customer information, products, prescriptions, orders, payments, deliveries, reviews, brands, suppliers, and store operations. The database design ensures data consistency and supports efficient business analysis.

3.2 ER Diagram

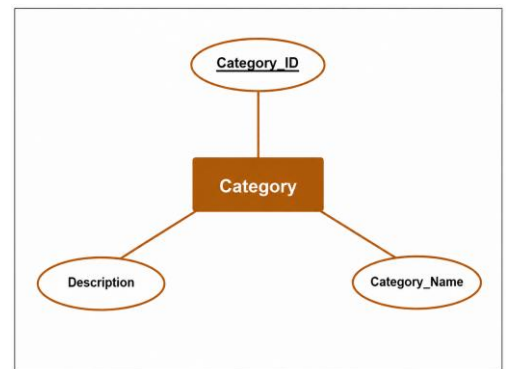
The Entity Relationship (ER) Diagram represents the logical structure of the Lenskart Analysis Database. It illustrates the entities, attributes, and relationships involved in the system.

Entity	Description
Customer	Stores customer information
Product	Stores eyewear product details
Category	Classifies products
Brand	Stores brand information
Supplier	Stores supplier information
Prescription	Stores eye prescription details
Cart	Stores shopping cart information
Order	Stores order information
Order Item	Stores products within orders
Payment	Stores payment details
Delivery	Stores delivery information
Review	Stores customer reviews
Store	Stores store information

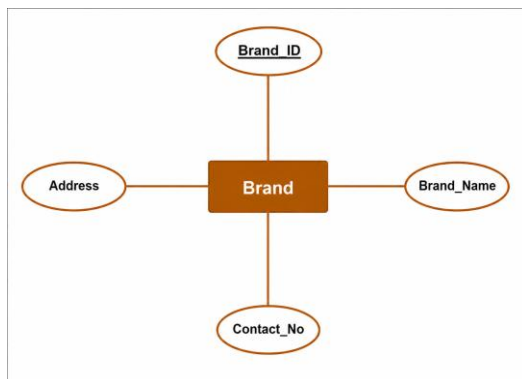
i. Customer



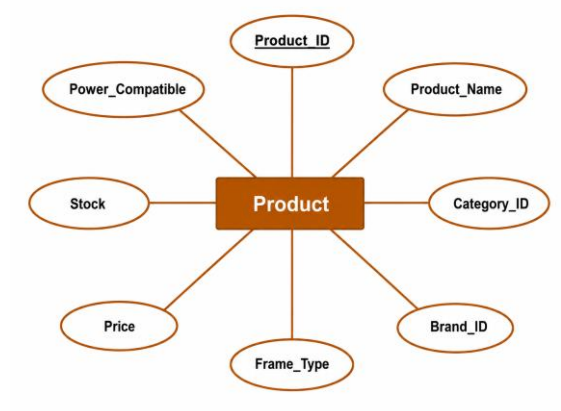
ii. Category



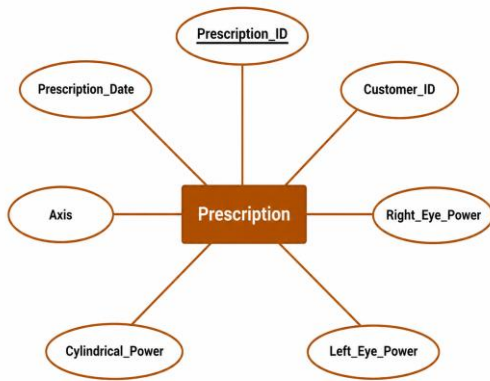
iii. Brand



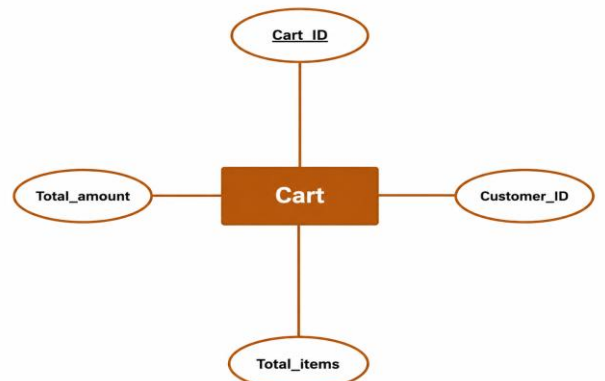
iv. Product



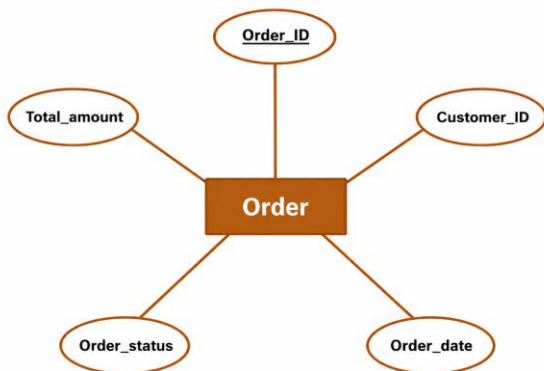
v. Prescription



vi. Cart



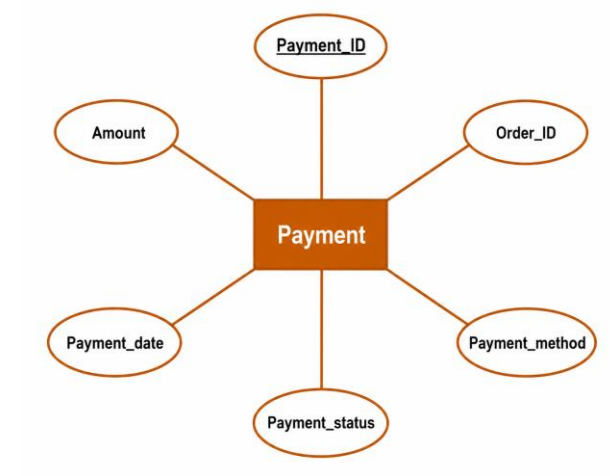
vii. Order



viii. Order Item (weak Entity)



xi. Payment



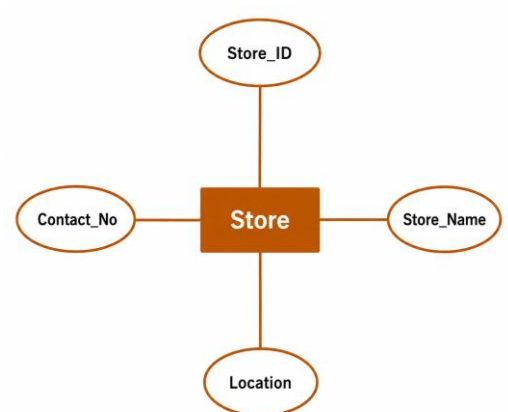
x. Review



xi. Delivery

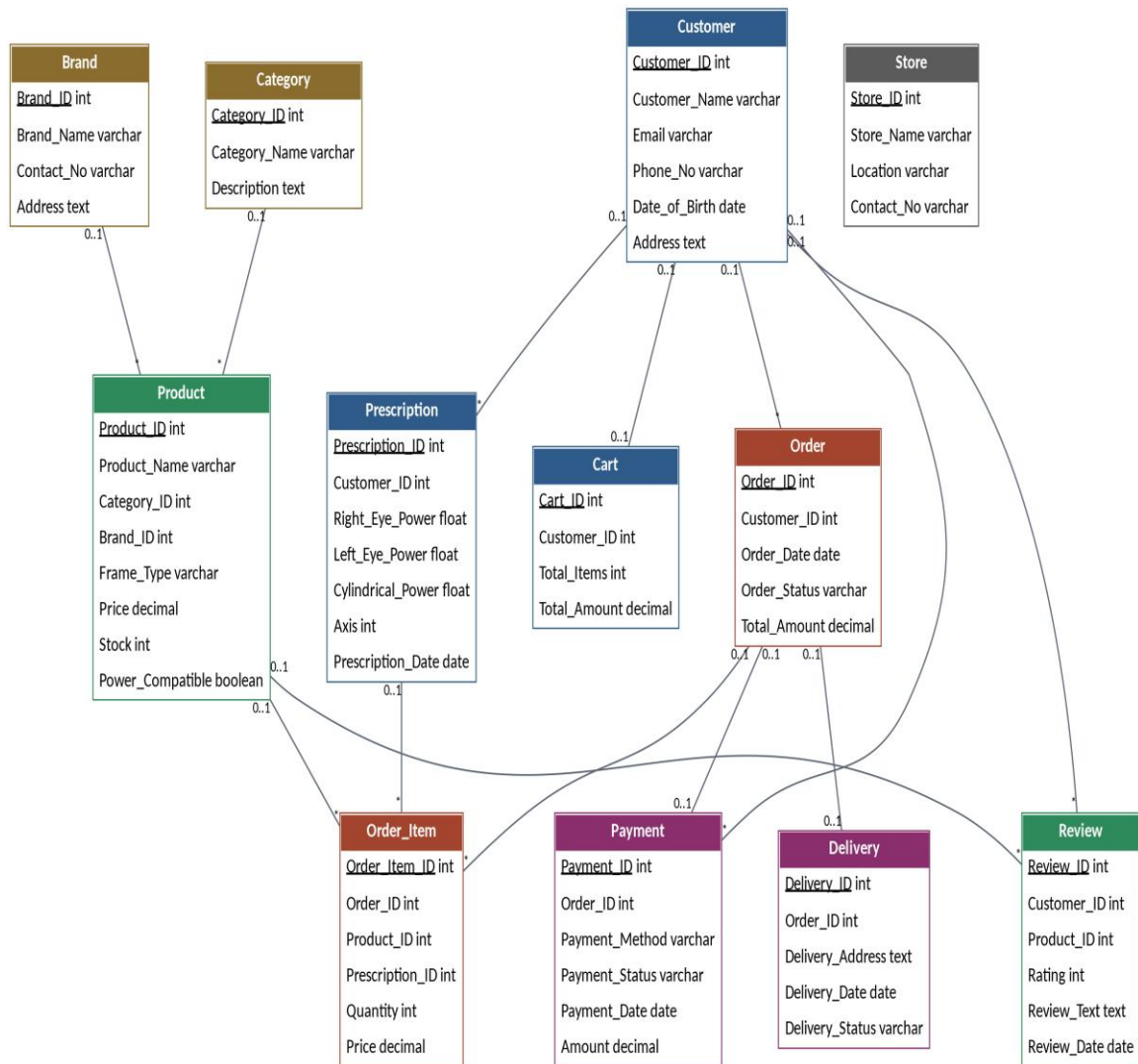


x. Store



3.3 Relational Schema

The relational schema shows the tables, attributes, primary keys, and foreign-key relationships derived from the ER model.



3.4 Overall ER Diagram

The table below summarizes the relationships between entities, along with their cardinality and participation constraints.

Entity 1	Relation	Entity 2	Cardinality	Participation
Customer	Owns	Cart	One-to-One	Partial / Total
Customer	Has	Prescription	One-to-Many	Partial / Total
Customer	Places	Order	One-to-Many	Partial / Total
Customer	Writes	Review	One-to-Many	Partial / Total
Category	Categorizes	Product	One-to-Many	Total / Partial
Brand	Belongs To	Product	One-to-Many	Partial / Total
Supplier	Supplies	Product	One-to-Many	Partial / Total
Order	Contains	Order Item	One-to-Many	Total / Total
Product	Included In	Order Item	One-to-Many	Partial / Total
Order	Generates	Payment	One-to-Many	Total / Total
Order	Generates	Delivery	One-to-Many	Total / Total
Product	Receives	Review	One-to-Many	Partial / Total
Store	Sells	Product	One-to-Many	Partial / Partial

